

Leaf+Pest Gate — Evaluation Results

TinyPestGate · 3-class leaf / others / pest · 96×96 RGB · INT8 TFLite

This gate replaces the binary leaf/not-leaf decision with a three-way split: **leaf** (continue to the disease CNN), **pest** (PlantVillage pest images), and **others** (non-leaf clutter). The student is the same depthwise-separable topology as TinyLeafGate with a 3-way head (28,883 parameters, 51.0 KB INT8). All test numbers below are copied from this folder's Colab capture *training_record.pdf* and from the class counts in *leaf_pest_others_dataset/*.

Sources used

Primary: *training_record.pdf* (Colab capture, 2026-08-06) — PyTorch FP32 test on page 8, INT8 test on page 12, confusion matrix on page 12. Dataset counts: *leaf_pest_others_dataset/{train,val,test}/{leaf,others,pest}*. Artifact: *aclis_pest_gate_96x_full_int8.tflite* (51.0 KB). Recipe notebook: *aclis_pest_gate_distill_qat.ipynb*.

1. Overall accuracy

Overall accuracy is measured on the held-out test split (n = 4,319). The deployed number is **clean INT8 = 95.58%**, identical to PyTorch FP32 on the same images (gap 0.00). That is above the notebook target of 90%. Camera-stress INT8 (same images with blur / JPEG / lighting) is 91.06% — still above target, but it shows the 3-class problem is harder under degradation than the binary leaf gate.

INT8 test accuracy	PyTorch FP32	Camera-stress INT8	Target
95.58%	95.58%	91.06%	≥ 90%
deployed model · n = 4,319	gap 0.00 · agreement 99.84%	blur + JPEG + underexposure	PASS

Metric	Value	Notes
Clean INT8 accuracy	0.9558 (95.58%)	tf.lite.Interpreter on full test set
Clean PyTorch FP32	0.9558 (95.58%)	Same split; Δ = 0.0000
Pred agreement INT8 ↔ PT	0.9984	Almost always the same argmax
Camera-stress INT8	0.9106 (91.06%)	Fixed camera-stress transform
Camera-stress FP32	0.9090 (90.90%)	Same degraded images, float student
Best EMA validation	0.9558	Student KD, 40 epochs
Teacher best val	0.9921	MobileNetV3-Small, 8 epochs (train-only)
TFLite size	51.0 KB	Disease 723 KB + gate 51 KB = 774 / 1024 KB (FITS)

Table 1. Overall accuracy for TinyPestGate. INT8 is the MCU artifact.

2. Per-class accuracy

Per-class accuracy is recall on the clean INT8 test set, taken both from the printed per-class line in the training record and from the confusion-matrix diagonal. The three classes are close to one another: pest is the strongest (96.13%), then leaf (95.57%), then others (95.38%). Camera-stress recall is listed beside the clean figure; off-diagonal stress counts were not printed in the record, so no stress confusion matrix is invented here.

Class	Test n	Correct (INT8)	INT8 recall	FP32 clean	INT8 stress
leaf (0)	1,943	1,857	95.57%	95.7%	91.5%
others (1)	1,730	1,650	95.38%	95.3%	89.0%
pest (2)	646	621	96.13%	96.0%	95.4%
Overall	4,319	4,128	95.58%	95.58%	91.06%

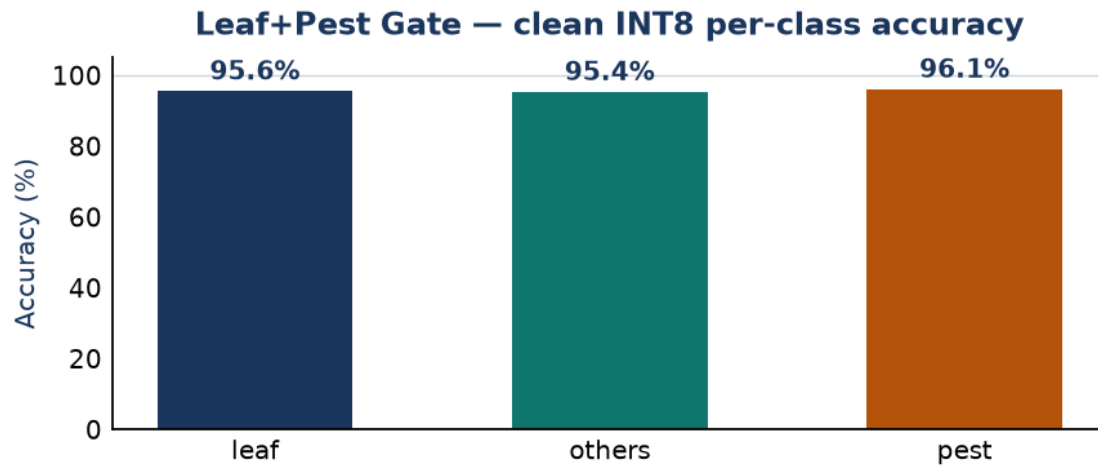


Figure 1. Clean INT8 per-class recall. Pest is the smallest class but the most accurate.

3. Confusion matrix

Printed in *training_record.pdf* section 8 as “Confusion matrix (rows=true, cols=pred)” after a full pass of the INT8 interpreter over all 4,319 test images. Dataset folder counts match the row totals: test/leaf = 1,943, test/others = 1,730, test/pest = 646.

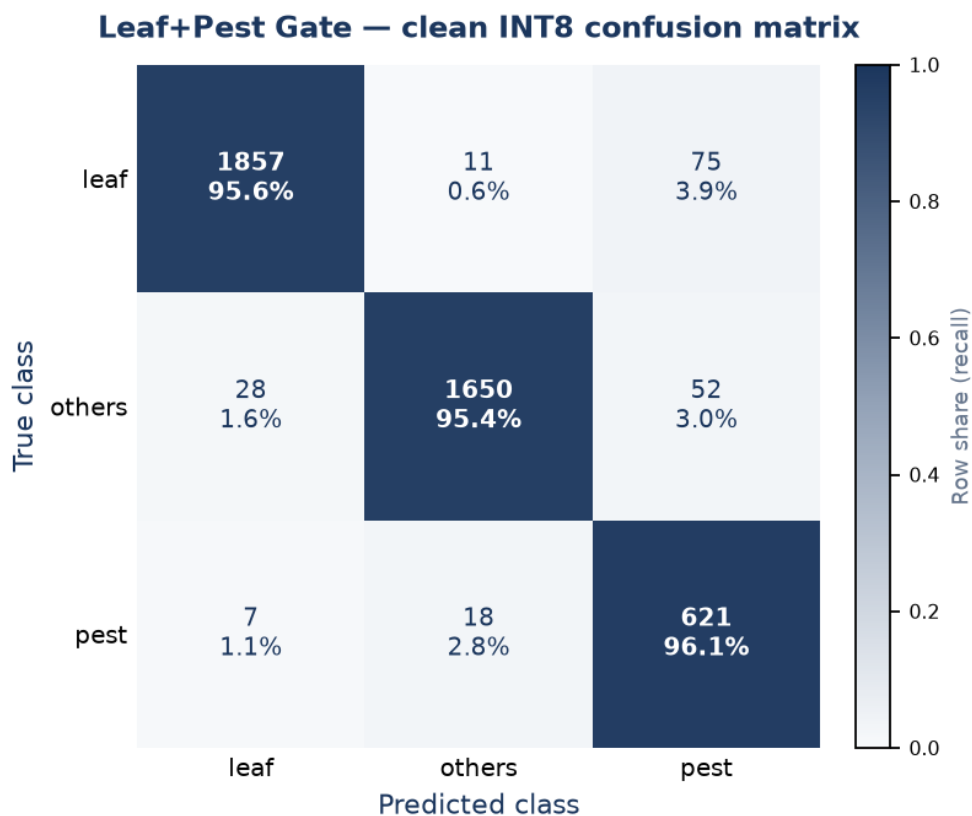


Figure 2. Leaf+Pest Gate confusion matrix, clean INT8 test ($n = 4,319$).

True \ Pred	leaf	others	pest	Row total	Recall
leaf	1,857	11	75	1,943	95.57%
others	28	1,650	52	1,730	95.38%
pest	7	18	621	646	96.13%
Column total	1,892	1,679	748	4,319	95.58%

Table 2. Numeric confusion matrix — pest-gate INT8, clean test.

How to read the matrix

Leaf errors (86 / 1,943). Almost all leaf mistakes go to *pest* (75), not to *others* (11). Visually that is expected: diseased or insect-damaged leaf tissue can look closer to the pest class than to random non-leaf clutter. A leaf→pest swap would skip the disease CNN; a leaf→others swap rejects the frame entirely.

Others errors (80 / 1,730). 52 are called pest and 28 are called leaf. False leaf is the cascade cost (disease CNN on junk); false pest is a wrong specialised path.

Pest errors (25 / 646). Pest is the most reliable class. The 7 pest→leaf swaps would incorrectly enter the disease CNN; the 18 pest→others swaps would drop a pest frame.

Train counts in this folder: leaf 9,065 / others 8,066 / pest 3,009. Pest is the minority class and was up-weighted (weight ≈ 2.23) during training.

4. Dataset split (from this folder)

Counts are file counts under *leaf_pest_others_dataset/*. leaf and others come from the binary leaf/not-leaf set; pest comes from PlantVillage pest images. Evaluation used full-frame Resize(96), matching firmware, not a center crop.

Split	leaf	others	pest	Total
train	9,065	8,066	3,009	20,140
val	1,942	1,728	644	4,314
test	1,943	1,730	646	4,319

Table 3. Class counts in *leaf_pest_others_dataset/*.